



**PERIODIC VALIDATION VALIDATION OF
FX SIMULATION MODEL FOR PFE
FOR ENTITY**

January 2026

Report **RVREP.1820** issued on **March 16, 2026**

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1. Executive Summary

This validation report assesses the adequacy of *FX Simulation Model For PFE* for use within the PFE framework. The review focuses on the FX **simulation** ¹ and the FX **calibration** methodology, including conceptual soundness, data integrity, implementation considerations, and performance testing requirements.

In particular, the validation investigated the following key areas:

- the **conceptual soundness** of the modelling framework, including the specification of FX spot dynamics, volatility modelling assumptions, and the interaction with other simulated risk factors;
- the **performance of the model**, through empirical analysis of FX return distributions and statistical diagnostics performed on historical data;
- the **implementation and numerical design** of the simulation methodology used to generate FX paths within the Monte Carlo framework;
- the **data integrity** of the market data inputs used in the calibration of the model;
- the **governance framework** surrounding the development, documentation, and maintenance of the model.

As part of the performance assessment, EMRM conducted several empirical diagnostic tests on historical FX return series, including:

- descriptive statistics and return distribution analysis;
- rolling volatility stability diagnostics;
- tail behaviour analysis using quantile–quantile plots and extreme value theory diagnostics;
- distributional tests applied to standardised residuals;
- backtesting of Value-at-Risk exceedance frequencies;
- regime clustering analysis using unsupervised statistical learning techniques.

The results of these analyses confirm that FX return distributions exhibit several stylised empirical features that are not fully captured by the modelling framework, including volatility clustering, heavy tails, and regime shifts in volatility levels. These findings highlight the limitations associated with the constant-volatility geometric Brownian motion specification adopted in the simulation model.

Nevertheless, the modelling approach remains broadly consistent with common practices in exposure simulation frameworks used for counterparty credit risk. The geometric Brownian motion specification provides a computationally efficient and tractable mechanism for generating FX scenarios within large-scale Monte Carlo simulations.

EMRM found that the implementation of the model is coherent with the documented methodology and that the data inputs used in the calibration process are clearly defined and traceable. The governance framework supporting the model includes formal documentation, defined model ownership, and periodic validation reviews.

Based on the analysis performed in this validation, EMRM identified several modelling simplifications related primarily

¹specification (Model document 1.2, Section 2.4)

to the representation of volatility dynamics and the modelling of implied volatility surfaces. These limitations are not uncommon in PFE simulation frameworks but may lead to a partial underestimation of tail risk in certain market conditions.

EMRM therefore recommends enhancing the model documentation to provide clearer justification of the modelling assumptions and to explicitly document the limitations associated with the simplified volatility modelling framework.

Based on the analysis presented in this report, the **Complexity Rating** of the model is assessed to be **Medium**, reflecting the relatively simple stochastic structure of the FX dynamics combined with the integration of the model within a multi-factor Monte Carlo simulation engine.

At the time of validation, the **Materiality Rating** of the model is assessed to be **High**, reflecting the importance of FX risk factors within the PFE framework and their potential impact on counterparty credit risk exposure calculations.

Combining the assessed complexity and materiality levels results in an overall **Model Risk Tier of High (HT)** under the EMRM model risk classification framework.

2. List of Findings

This section lists findings identified during the independent review of the FX simulation model ² and its calibration approach.

2.1. List of Existing Findings

Finding: Number of Monte Carlo Simulations

Finding		Recommendation	
EMRM found that the number of Monte Carlo Simulations in place is not sufficient to ensure convergence to drivers assumed distributional properties.		EMRM recommends RAG increasing the number of Monte Carlo simulations in order to reduce current simulation sampling error which is considered to be material.	
Risk Rating	Medium	Owner: Model Owner	Date: Open

"Percentile.INC" estimator use underestimating the extreme quantiles

Finding		Recommendation	
EMRM found that the current use of the "PERCENTILE.INC" estimator potentially leads to underestimation of extreme quantiles.		EMRM recommends RAG investigating QuilC's capabilities and implementing the "CENTRAL" estimator during PFE quantile calculation.	
Risk Rating	Low	Owner: Model Owner	Date: Closed

Cross-Currency not modelled

Finding		Recommendation	
EMRM assessment demonstrates the importance of including cross-currency spread when modelling FX rates. The currency setting could underestimate when the spread widens.		EMRM recommends RAG to enhance the modelling by including the cross-currency spread.	
Risk Rating	Medium	Owner: Model Owner	Date: Open

²(Model document 1.2, Section 2.4)

Missing mean-reversion control

Finding		Recommendation	
EMRM found the potential future 95th percentile to be subject to decreasing or reverse growth for high volatilities. Mean reversion could help controlling the percentile behaviour, especially that arguments for mean reversion in case of managed (a.k.a. pegged) currencies were highlighted by EMRM.		EMRM recommends RAG to investigate the need for inclusion of mean-reversion to control the PFE path behaviour.	
Risk Rating	Low	Owner: Model Owner	Date: To Be Confirmed

Mode Documentation

Finding		Recommendation	
EMRM notes that the model documentation describes in a comprehensive manner model's methodology. Furthermore calibration procedure is well outlined in a separate calibration document. Furthermore calibration procedure is well outlined in a separate calibration document. However EMRM found that both documents cited above are not in line with new risk model template. Additionally, model input data as well as model implementation and testing are not currently discussed.		EMRM recommends RAG moving model documentation to the new Risk template standard. This template will cover model input data and implementation along with other model components and testing.	
Risk Rating	Medium	Owner: Model Owner	Date: Closed

3. Assumptions and Limitations of the FX Simulation Model

3.1. Assumptions of the Foreign Exchange Simulation Model

The foreign exchange (FX) rates in the PFE simulation framework are modelled using a geometric Brownian motion (GBM). The specification of the process and its calibration rely on a number of structural and data-related assumptions. These assumptions are detailed below.

3.1.1. Lognormal Dynamics of FX Rates

The FX rate $Y_k^{FX}(t)$ for currency pair k is assumed to follow a geometric Brownian motion process:

$$Y_k^{FX}(t) = g_k^{FX}(t) \exp\left(-\frac{1}{2}\sigma_k^2 t + \sigma_k W_k(t)\right) \quad (1)$$

where $W_k(t)$ is a standard Brownian motion and σ_k is the FX volatility.

This specification implies that FX rates are lognormally distributed. Consequently, the model assumes:

- FX rates remain strictly positive at all times.
- Log-returns of FX rates are normally distributed.
- The diffusion term is continuous with no jumps or discontinuities.

3.1.2. Constant Volatility

The FX volatility σ_k is assumed to be constant over time:

$$\sigma_k(t) = \sigma_k \quad (2)$$

This implies that the model does not incorporate stochastic volatility effects, volatility clustering, or volatility regime changes.

3.2. Deterministic Drift Derived from Interest Rate Curves

The deterministic mean function $g_k^{FX}(t)$ is derived from the domestic and foreign interest rate curves:

$$g_k^{FX}(t) = Y_k^{FX}(0) \frac{DF_f(0, t)}{DF_d(0, t)} \quad (3)$$

where $DF_d(0, t)$ i.e. $DF_{USD}(0, t)$ and $DF_f(0, t)$ denote the domestic and foreign discount factors respectively.

This formulation assumes:

- Covered interest parity holds exactly.
- The expected FX rate under the domestic risk-neutral measure equals the FX forward curve implied by the two interest rate curves.
- Interest rate curves used in the drift calculation are deterministic inputs to the FX simulation.

3.2.1. Historical Volatility Estimation

The volatility parameter σ_k is estimated using historical FX returns over a three-year observation window. Specifically, the model assumes that the volatility can be estimated from the standard deviation of historical log returns:

$$\sigma_k = \frac{\text{Std} \left(\ln \left(\frac{Y_k^{FX}(t_{i+5})}{Y_k^{FX}(t_i)} \right) \right)}{\sqrt{5}} \quad (4)$$

This implies the following assumptions:

- Historical volatility is a suitable proxy for future FX volatility.
- The historical window of three years is representative of future market conditions.
- The statistical properties of FX returns are stationary over the estimation period.

3.2.2. Gaussian Innovations

The model assumes that the innovations driving the FX process are Gaussian:

$$Z \sim \mathcal{N}(0, 1) \quad (5)$$

This assumption implies:

- FX returns exhibit no excess kurtosis relative to the normal distribution.
- Tail risk is entirely determined by the variance parameter.
- Extreme market movements are not explicitly modelled.

3.2.3. Correlation with Other Risk Factors

FX rates are simulated jointly with other market risk factors such as interest rates and credit spreads using a correlated Brownian motion framework. The correlation structure is captured through a covariance matrix $C(\Delta t)$.

This implies the following assumptions:

- Linear correlations between risk factors are stable over time.
- Dependence between risk factors can be fully described through a Gaussian correlation structure.
- Higher-order dependence structures (such as tail dependence) are not modelled.

3.2.4. Market Data Assumptions

The calibration of the FX model relies on the following market data inputs:

- a three-year time series of historical FX spot rates;
- the current FX spot rate at the simulation start date;
- domestic and foreign discount curves.

The model therefore assumes that:

- the market data used for calibration is accurate and free from structural inconsistencies;
- missing observations in the historical time series have been appropriately treated through data cleansing procedures.

3.2.5. Discrete-Time Simulation

The continuous-time stochastic differential equation is simulated using a discrete-time approximation:

$$\ln \left(\frac{Y_k^{FX}(t + \Delta t)}{Y_k^{FX}(t)} \right) = \left(b_k - \frac{1}{2} \sigma_k^2 \right) \Delta t + \sigma_k \sqrt{\Delta t} Z \quad (6)$$

This discretisation assumes that the simulation time step Δt is sufficiently small for the Euler approximation to remain accurate.

3.3. Assumptions of the Implied Volatility Model

The FX modelling framework used in the PFE engine consists of : a stochastic model describing the evolution of implied volatility surfaces used in the valuation of option positions. The specification of this process relies on several modelling assumptions described below.

3.3.1. Implied Volatility as a Risk Factor

The framework assumes that implied volatility is itself a stochastic risk factor that must be simulated in order to capture the exposure of option positions.

This assumption reflects the fact that option prices depend on both the level of the underlying asset and the level of implied volatility.

3.3.2. Proportional Evolution of the Volatility Surface

The entire implied volatility surface $S_k(\Psi, t)$ is assumed to evolve proportionally through a single volatility shift factor $Y_k^{IV}(t)$:

$$S_k(\Psi_i, t) = Y_k^{IV}(t) S_k(\Psi_i, 0)$$

This assumption implies that:

- all points of the volatility surface move proportionally;
- the relative shape of the surface remains constant;
- the smile and skew structure are preserved through time.

3.3.3. Mean-Reverting Dynamics of the Volatility Shift Factor

The volatility shift factor follows a mean-reverting stochastic process driven by a Gaussian innovation:

$$dX_k^{IV}(t) = -\lambda_k X_k^{IV}(t) dt + \sigma_k dW_k(t)$$

The shift factor itself is obtained through an exponential transform:

$$Y_k^{IV}(t) = \exp \left(X_k^{IV}(t) - \frac{1}{2} v_k^2(t) \right)$$

This specification ensures that:

- implied volatilities remain positive;
- the expected value of the volatility surface remains equal to its current level.

3.3.4. Single-Factor Representation of the Volatility Surface

The dynamics of the volatility surface are represented by a single stochastic factor.

This implies that the evolution of a high-dimensional volatility surface is approximated by a single driving process.

3.3.5. Global Calibration Parameters

Due to the instability of parameter estimation, the volatility and mean reversion parameters of the volatility shift factor are set to global constant values across tenors and currencies.

This assumes that:

- a common parameterisation provides a sufficiently accurate representation of volatility dynamics;
- differences in volatility behaviour across maturities and instruments are negligible for exposure simulation purposes.

3.3.6. Independence of Volatility Factors

Volatility shift factors are assumed to be independent from other risk factors and from each other.

This implies that:

- implied volatility changes are not correlated with spot FX movements;
- volatility shocks are independent across different markets.

3.4. Limitations of the Foreign Exchange Simulation Model

Although the FX simulation model provides a tractable and computationally efficient framework for generating scenarios in the PFE engine, several limitations arise from the modelling choices and calibration methodology. These limitations may affect the accuracy of simulated FX dynamics and the resulting exposure profiles.

3.4.1. Constant Volatility Assumption

The model assumes that FX volatility σ_k is constant over time. In practice, FX markets exhibit several stylised features that are not captured by this assumption:

- volatility clustering;
- regime shifts in volatility levels;
- stochastic volatility dynamics;
- implied volatility smiles and skews observed in FX options markets.

As a result, the model may underestimate the probability of large FX movements and may not accurately represent the distribution of FX rates in stressed market conditions.

3.4.2. Absence of Jumps in FX Dynamics

The FX process is modelled as a continuous diffusion driven by Brownian motion. This specification excludes the possibility of discontinuous price movements.

However, FX markets may experience sudden jumps due to macroeconomic announcements, central bank interventions, or geopolitical events. The absence of jump components may therefore lead to an underestimation of tail risk in the simulated FX paths.

3.4.3. Use of Historical Volatility Instead of Implied Volatility

The volatility parameter is calibrated using historical FX returns over a three-year window. This approach presents several limitations:

- historical volatility may not reflect current market expectations;
- implied volatility embedded in FX options is ignored;
- the model may not respond quickly to changes in market volatility regimes.

Consequently, the simulated FX distribution may diverge from market-implied risk perceptions.

3.4.4. Simplified Interaction Between FX and Interest Rates

The drift of the FX process is derived from domestic and foreign discount curves through the covered interest parity relationship. While this ensures arbitrage consistency at the initial time, the model assumes that interest rate curves entering the FX drift remain deterministic.

In reality, FX rates and interest rates evolve jointly and exhibit stochastic interactions. The simplified representation may therefore underestimate the dependency between FX dynamics and interest rate movements.

3.4.5. Gaussian Distribution of Innovations

The model assumes that the innovations driving the FX process follow a standard normal distribution. Empirical FX returns, however, are known to exhibit:

- excess kurtosis;
- fat tails;
- asymmetric return distributions.

The Gaussian assumption therefore underestimates the probability of extreme FX moves and may lead to an underestimation of tail exposures in the PFE distribution.

3.4.6. Static Correlation Structure

The dependence between FX rates and other simulated risk factors is modelled through a fixed correlation matrix. This approach assumes that correlations between risk factors are constant over time.

In practice, correlations between FX, interest rates, and credit spreads tend to increase during market stress. The use of static correlations may therefore underestimate wrong-way risk and systemic market movements.

3.4.7. Limited Historical Data Window

The calibration of the FX volatility relies on approximately three years of historical observations. This relatively short time window may not capture rare but impactful market events such as financial crises or currency shocks. As a consequence, the calibrated volatility may underestimate long-term FX risk.

3.4.8. Model Suitability for Complex FX Derivatives

The geometric Brownian motion specification is well suited for simulating spot FX exposures associated with linear products such as FX forwards or cross-currency swaps. However, it may not adequately capture the risk profile of portfolios containing nonlinear FX derivatives such as:

- FX options;
- barrier options;
- structured FX products.

In such cases, the model may fail to reproduce the dynamics implied by the volatility surface observed in the FX options market.

3.4.9. Potential Underestimation of Tail Exposure in PFE

Due to the combination of constant volatility, Gaussian innovations, and the absence of jumps, the model may underestimate extreme FX movements. This limitation may lead to an underestimation of tail exposures and consequently

of high quantiles of the exposure distribution, such as the 97.5% or 99% PFE levels.

3.5. Limitations of the Implied Volatility Modelling Framework

In order to capture the exposure of portfolios containing options, the PFE simulation framework models the evolution of implied volatility surfaces as additional stochastic risk factors. While this approach allows option sensitivities to volatility to be incorporated into exposure simulations, the modelling framework remains simplified and introduces several limitations.

3.5.1. Single-Factor Representation of the Volatility Surface

The evolution of the implied volatility surface $S_k(\Psi, t)$ is represented by a single multiplicative shift factor $Y_k^{IV}(t)$:

$$S_k(\Psi_i, t) = Y_k^{IV}(t) S_k(\Psi_i, 0) \quad (7)$$

This implies that all points of the volatility surface move proportionally over time.

In practice, implied volatility surfaces exhibit complex dynamics involving:

- changes in the slope of the volatility smile;
- twists in the term structure of volatility;
- localized deformations ("bumps") at specific strikes or maturities.

Because the model assumes a common multiplicative shift across all surface points, it cannot reproduce these richer dynamics. Consequently, the simulated volatility surfaces may not accurately reflect observed market behaviour.

3.5.2. Absence of Smile and Skew Dynamics

FX volatility surfaces typically exhibit pronounced smile and skew structures that evolve over time. In the present framework, the proportional scaling assumption implies that the relative shape of the volatility surface remains constant.

Therefore:

- the skew of the volatility surface is assumed to remain unchanged;
- the curvature of the volatility smile cannot evolve independently;
- the model cannot reproduce smile dynamics observed in stressed markets.

This limitation may lead to inaccuracies in the valuation of path-dependent FX options and other nonlinear derivatives during the simulation.

3.5.3. Mean-Reverting Lognormal Volatility Shift Factor

The volatility shift factor $Y_k^{IV}(t)$ follows a lognormal process driven by a mean-reverting Ornstein-Uhlenbeck driver:

$$Y_k^{IV}(t) = g_k^{IV}(t) \exp \left(X_k^{IV}(t) - \frac{1}{2} v_k^2(t) \right) \quad (8)$$

with

$$dX_k^{IV}(t) = -\lambda_k X_k^{IV}(t)dt + \sigma_k dW_k(t) \quad (9)$$

In the implemented specification, the deterministic component satisfies

$$g_k^{IV}(t) = 1 \quad (10)$$

which implies that the expected value of each volatility surface point remains equal to its current level.

While this specification ensures positivity of implied volatilities, it does not reflect the empirical behaviour of volatility surfaces, which often display persistent trends and regime shifts.

3.5.4. Use of Global Parameters

The calibration procedure indicates that the estimation of the volatility and mean reversion parameters of the shift factor is unstable when performed independently for each tenor. As a result, the model adopts fixed global parameter values:

$$\sigma = 0.35, \quad \lambda = 0.3 \quad (11)$$

Similarly, conservative global values are used in other contexts (e.g. swaption volatility modelling).

While this approach improves stability, it introduces several limitations:

- the parameters are not calibrated to the specific dynamics of each volatility surface;
- the model may not reflect differences across currencies or maturities;
- the calibration may become disconnected from observed market dynamics.

3.5.5. Limited Calibration Accuracy

The model documentation explicitly acknowledges that the simplified representation of the volatility surface limits the ability to fit historical volatility dynamics.

Since a single factor is used to represent the evolution of a high-dimensional volatility surface, only a coarse approximation of historical volatility behaviour can be achieved.

3.5.6. Independence of Volatility Shift Factors

The volatility shift factors associated with different surfaces are assumed to be uncorrelated with each other and with other risk factors:

- volatility shift factors are not correlated across currencies;
- they are not correlated with interest rates or FX spot dynamics.

Empirical evidence suggests that volatility tends to increase during periods of market stress and often exhibits strong correlation with other market variables. Ignoring these dependencies may therefore underestimate joint stress scenarios.

3.5.7. Potential Impact on Exposure Profiles

Because implied volatility is a key determinant of option prices, the simplified volatility surface dynamics may affect the simulated exposure of portfolios containing options. In particular:

- the model may underestimate exposure during volatility spikes;
- changes in option moneyness may interact with an unrealistic volatility surface;
- nonlinear products may exhibit exposure profiles that differ from those implied by market-consistent volatility dynamics.

Consequently, the simplified implied volatility modelling framework may introduce model risk when evaluating PFE for portfolios with significant optionality.

3.6. Model Risk Assessment for the FX Simulation Model

The following table summarises the main modelling assumptions underlying the FX simulation framework, the resulting limitations, and their potential impact on the estimation of Potential Future Exposure (PFE).

Model Risk Assessment for the FX Simulation Model

Model Assumption	Model Limitation	Potential Impact on PFE	Severity
FX spot follows geometric Brownian motion	The model assumes continuous lognormal dynamics and excludes jumps or discontinuous movements.	Sudden FX shocks due to macroeconomic announcements or geopolitical events are not captured, potentially underestimating tail exposures.	Medium
Constant FX volatility	Volatility is assumed constant over time and does not incorporate stochastic volatility or regime changes.	Exposure distributions may underestimate volatility spikes and therefore underestimate high quantiles of PFE.	Medium
Historical volatility calibration	Volatility is estimated from historical returns over a fixed observation window.	Historical volatility may differ from current market expectations reflected in implied volatility, potentially misrepresenting future risk.	Medium
Gaussian innovations	The model assumes normally distributed shocks.	Fat tails and excess kurtosis observed in FX markets are not captured, leading to possible underestimation of extreme FX movements.	Medium
Deterministic drift derived from interest rate curves	The FX drift is determined by current interest rate curves and does not incorporate stochastic interaction with future interest rate movements.	Interaction between FX and interest rate dynamics may be simplified, potentially affecting exposure profiles of cross-currency instruments.	Low
Static correlation matrix	Correlations between FX and other risk factors are assumed constant.	Correlations tend to increase during periods of market stress, which may lead to underestimation of joint extreme scenarios.	Medium
Implied volatility surface represented by a single factor	The entire volatility surface moves proportionally according to a single shift factor.	Changes in smile or skew dynamics cannot be captured, reducing accuracy of option valuation under stress scenarios.	High
Proportional volatility surface evolution	Relative shape of the volatility surface remains constant through time.	Twists, bumps, and slope changes observed in market volatility surfaces cannot be reproduced.	High

Model Assumption	Model Limitation	Potential Impact on PFE	Severity
Volatility shift factor follows a simple mean-reverting process	The stochastic driver may not fully represent complex volatility dynamics observed in option markets.	Volatility spikes or prolonged volatility regimes may not be accurately simulated, affecting option exposures.	Medium
Global parameter calibration for volatility shift factor	Model parameters are fixed globally rather than calibrated independently for each volatility surface.	Local market characteristics and maturity-specific volatility behaviour may not be captured.	Medium
Independence of volatility factors from other risk factors	Volatility shocks are assumed uncorrelated with FX spot movements and other market factors.	Empirical relationships such as spot-volatility dependence are ignored, which may underestimate stress exposures.	Medium
Single-factor reduction of high-dimensional volatility surface	A high-dimensional volatility surface is approximated by a single stochastic driver.	The dimensionality reduction may limit the model's ability to reproduce realistic volatility surface dynamics.	High

4. FX Simulation Model Description

Foreign exchange (FX) rates are modelled explicitly in the simulation framework as stochastic risk factors. The model represents FX rates relative to a reporting currency, which is assumed to be USD. Consequently, exposures denominated in USD and not dependent on FX rates carry no FX risk.

The FX modelling framework is designed to be consistent with the stochastic evolution of other market risk factors simulated in the system, including interest rates, credit spreads, inflation indices and volatility factors. FX rates are simulated using a geometric Brownian motion specification whose drift is determined by a deterministic mean function derived from interest rate differentials between the domestic and foreign currencies.

The FX dynamics are calibrated using historical time series of FX spot rates together with the current interest rate curves for the relevant currencies. For instruments with optionality, such as FX options, an additional model for the implied volatility surface is required. This surface is represented using a simplified stochastic structure driven by a volatility shift factor.

4.1. FX Rate Dynamics

Let $Y_k^{FX}(t)$ denote the FX rate of currency k expressed as USD per unit of foreign currency.

The stochastic driver $X_k^{FX}(t)$ governing the FX process follows a Brownian motion process of the form

$$dX_k^{FX}(t) = \sigma_k dW_k(t) \quad (12)$$

where

- $W_k(t)$ is a standard Brownian motion
- σ_k denotes the FX volatility parameter
- k indexes the currency pair

The FX rate is obtained through the transformation

$$Y_k^{FX}(t) = g_k^{FX}(t) \exp \left(X_k^{FX}(t) - \frac{1}{2} \nu_k^2(t) \right) \quad (13)$$

where

- $g_k^{FX}(t)$ is the deterministic mean function
- $\nu_k^2(t)$ denotes the variance of the stochastic driver.

Under this formulation, the FX rate follows a geometric Brownian motion with drift determined by the deterministic function $g_k^{FX}(t)$. The expected value of the FX rate therefore satisfies

$$\mathbb{E}[Y_k^{FX}(t)] = g_k^{FX}(t) \quad (14)$$

which ensures that the expected FX level equals the forward FX curve implied by market data.

4.1.1. Mean Function of FX Rates

The deterministic mean function is derived from the interest rate curves of the domestic and foreign currencies.

Let

- $DF_d(0, t)$ denote the domestic discount factor
- $DF_f(0, t)$ denote the foreign discount factor
- $Y_k^{FX}(0)$ denote the current spot FX rate

The mean function is then defined as

$$g_k^{FX}(t) = Y_k^{FX}(0) \frac{DF_f(0, t)}{DF_d(0, t)} \quad (15)$$

This specification ensures that the drift of the FX process is consistent with the FX forward curve implied by interest rate parity.

4.2. Implied Volatility Modelling

4.2.1. Volatility Surface Representation

The valuation of option positions depends on the implied volatility of the underlying FX rate. Implied volatility is therefore treated as an additional risk factor in the simulation framework.

In general, the implied volatility of the k -th underlying is represented as a surface

$$S_k(\Psi, t) \quad (16)$$

where Ψ denotes the attributes defining a specific option contract.

For FX options, the vector Ψ typically contains

- option expiry
- option delta (or equivalently moneyness)

The volatility surface is specified on a discrete grid

$$\Psi_1, \Psi_2, \dots, \Psi_p$$

and volatility values for intermediate points are obtained through interpolation.

4.2.2. Simplified Evolution of the Volatility Surface

In principle, each grid point on the volatility surface could evolve as an independent stochastic risk factor. However, enforcing no-arbitrage conditions across the entire surface would impose complex constraints between grid points.

To maintain tractability in the simulation framework, a simplified modelling approach is adopted in which the entire surface evolves proportionally.

Specifically, for any two grid points Ψ_i and Ψ_j , the following relationship holds:

$$\frac{S_k(\Psi_i, t)}{S_k(\Psi_i, 0)} = \frac{S_k(\Psi_j, t)}{S_k(\Psi_j, 0)} \quad (17)$$

This implies that the evolution of the entire volatility surface can be represented by a single volatility shift factor $Y_k^{IV}(t)$.

Consequently, the volatility surface evolves according to

$$S_k(\Psi_i, t) = Y_k^{IV}(t) S_k(\Psi_i, 0) \quad (18)$$

for all grid points i .

This modelling choice preserves the no-arbitrage structure of the surface and reflects the empirical observation that implied volatilities tend to move in the same direction across maturities and deltas.

4.2.3. Volatility Shift Factor Dynamics

The volatility shift factor $Y_k^{IV}(t)$ is modelled using a lognormal process driven by a stochastic factor $X_k^{IV}(t)$:

$$Y_k^{IV}(t) = g_k^{IV}(t) \exp\left(X_k^{IV}(t) - \frac{1}{2}\nu_k^2(t)\right) \quad (19)$$

In this specification

$$g_k^{IV}(t) = 1 \quad (20)$$

which implies that the expected value of the volatility surface remains equal to its initial level.

The driver process follows an Ornstein–Uhlenbeck specification

$$dX_k^{IV}(t) = -\lambda_k X_k^{IV}(t)dt + \sigma_k dW_k(t) \quad (21)$$

where

- λ_k denotes the mean reversion speed
- σ_k denotes the volatility of the volatility shift factor.

4.3. Model Calibration

4.3.1. Calibration of FX Dynamics

The FX model parameters are calibrated using

- historical time series of FX spot rates
- current domestic interest rate curves
- current foreign interest rate curves

The dataset used in calibration includes

- a three-year historical series of FX spot rates $Y_k^{FX}(t_i)$
- discount factors $DF_d(0, t)$ and $DF_f(0, t)$
- the current spot FX rate $Y_k^{FX}(0)$

The mean function $g_k^{FX}(t)$ is determined directly from the interest rate curves. The volatility parameter σ_k is estimated from historical FX returns.

4.3.2. Calibration of FX Volatility of Volatility

For time series indexed by observation times t_i , the standard deviation of volatility shift increments is defined as

$$\theta_k(m) = \text{std} \left(\log(Y_k(t_{i+m})) - \left(1 - \frac{\mu_k}{52}\right) \log(Y_k(t_i)) \right) \quad (22)$$

where m represents the time step and μ_k denotes the drift parameter.

A grid of empirical standard deviations is computed across several maturities and tenors. Model parameters are then obtained by solving the optimisation problem

$$\min_{\lambda_k, \sigma_k} \sum_{m \in M} |\nu_k(m; \lambda, \sigma) - \theta_k(m)|^2 \quad (23)$$

Due to instability observed in this optimisation procedure, conservative constant parameters are adopted.

For FX volatility dynamics the parameters are set to

$$\sigma = 0.35, \quad \lambda = 0.3$$

4.3.3. Calibration of Swaption Volatility

For interest rate swaptions, a similar calibration approach is applied using historical volatility estimates. Conservative parameter values are adopted following backtesting results:

$$\sigma = 0.3, \quad \lambda = 0.5$$

These parameters represent the volatility and mean reversion of the volatility shift factor for the swaption volatility surface.

4.3.4. Global Parameters for Volatility Factors

Given the simplified structure of the volatility surface model, a single volatility shift factor represents the behaviour of multiple volatility surface points.

Consequently, it is not possible to achieve a perfect fit to historical data across all maturities simultaneously. For robustness and numerical stability, global constant parameters are therefore used for the volatility shift factor across currencies.

4.3.5. Calibration of Correlations

Correlations between risk factors are calibrated from the same historical dataset used for volatility calibration.

Residuals from the stochastic processes of each risk factor are computed and the corresponding time series are aligned. Pairwise correlations are then estimated from these residual series.

Specific modelling assumptions include

- interest rate correlations are derived consistently with relative volatilities
- inflation indices are correlated with other risk factors only through their drift components
- volatility shift factors are assumed to be uncorrelated with each other and with all other risk factors

5. Data Integrity

This section assesses the integrity, reliability, and appropriateness of the data used in the FX simulation model. The assessment focuses on the data sources, data processing procedures, data quality controls, and potential limitations that may affect model calibration and simulation outputs.

5.1. Data Sources

The calibration of the FX simulation model relies on both market data and historical time series observations. The primary data sources used in the model include:

- Historical time series of FX spot rates for each currency pair included in the simulation framework.
- Current FX spot rates used as the initial conditions for the simulation.
- Domestic and foreign interest rate curves used to derive FX forward rates through interest rate parity.
- Historical volatility data used in the calibration of volatility parameters.
- Market implied volatility surfaces for FX options where applicable.

Historical FX spot data is typically obtained from internal market data repositories or external market data providers. Interest rate curves are derived from market instruments such as swaps, deposits, and government bonds, and are constructed using the institution's standard curve-building methodology.

5.2. Data Used for Model Calibration

The calibration of the FX model uses a historical dataset consisting of approximately three years of FX spot observations. These time series are used to estimate the volatility parameters governing the stochastic driver of the FX rate dynamics.

For each currency pair k , the dataset consists of:

- Historical observations of the FX spot rate $Y_k^{FX}(t_i)$
- The current FX spot rate $Y_k^{FX}(0)$
- Domestic and foreign discount factors $DF_d(0, t)$ and $DF_f(0, t)$

The deterministic drift component of the FX process is derived from the current interest rate curves, ensuring consistency with the forward FX curve implied by market data.

5.3. Data Pre-processing

Prior to calibration, the raw market data undergoes several preprocessing steps to ensure consistency and usability.

These steps include:

- Alignment of time series across different market variables to ensure consistent observation dates.
- Removal of non-trading days and correction for missing observations.
- Outlier detection and filtering of anomalous observations.
- Interpolation or smoothing of interest rate curves where required.

- Conversion of FX quotes into a consistent quotation convention (e.g., USD per unit of foreign currency).

These preprocessing steps ensure that the data used for calibration is internally consistent and suitable for statistical estimation.

5.4. Data Quality Controls

Several quality checks are applied to ensure the reliability of the input data used in the model.

These checks include:

- Validation of FX spot rates against independent market data sources.
- Verification of discount factors derived from interest rate curves.
- Detection of missing or irregular observations in historical time series.
- Consistency checks between spot FX rates and implied forward FX rates derived from interest rate curves.

In cases where anomalies are detected, the affected data points are either corrected using validated market data sources or excluded from the calibration dataset.

5.5. Data Governance

Market data used in the model is sourced from controlled internal data repositories which follow established governance procedures. These procedures include regular updates, data validation protocols, and audit trails documenting the origin and transformation of market data.

Access to the data sources used in the model is restricted and managed according to internal data governance policies. This ensures traceability and reproducibility of the model calibration process.

5.6. Limitations of the Data

Despite the quality controls in place, several limitations related to the data should be acknowledged.

First, the calibration relies on historical data which may not fully capture future market conditions or structural changes in FX markets.

Second, the availability of long and consistent time series may vary across currency pairs, particularly for emerging market currencies.

Third, implied volatility surfaces may be incomplete or sparsely quoted for certain maturities or deltas, requiring interpolation or extrapolation techniques.

Finally, historical correlations between risk factors are estimated from past data and may not remain stable during stressed market environments.

5.7. Overall Assessment

Based on the review performed, the data used in the calibration of the FX simulation model is considered appropriate for the intended application. The data sources are reliable, preprocessing procedures ensure consistency, and data quality controls are in place to mitigate potential data errors.

While some limitations related to historical data and market coverage remain, these limitations are common in financial modelling applications and are not expected to materially affect the reliability of the model outputs for exposure simulation purposes.

6. Conceptual Soundness

This section assesses the conceptual soundness of the FX simulation model used in the exposure simulation framework. The assessment focuses on the theoretical consistency of the modelling approach, the economic justification of the assumptions, and the appropriateness of the modelling choices given the intended use of the model.

6.1. Economic Rationale of the FX Model

The FX rate dynamics are modelled using a geometric Brownian motion representation. This choice is widely adopted in quantitative finance due to its tractability and its ability to ensure positivity of exchange rates. The model assumes that the FX rate evolves according to a lognormal process whose drift is determined by interest rate differentials between the domestic and foreign currencies.

The deterministic mean function is derived from interest rate parity:

$$g_k^{FX}(t) = Y_k^{FX}(0) \frac{DF_f(0, t)}{DF_d(0, t)} \quad (24)$$

This formulation ensures that the expected FX rate under the simulation framework follows the forward FX curve implied by the current interest rate term structures. As a result, the model is consistent with no-arbitrage conditions in the FX market.

The stochastic component of the FX dynamics captures short-term fluctuations around this deterministic forward path and is calibrated using historical FX return data.

6.2. Consistency with Market Observables

The model structure ensures consistency with the main observable quantities in the FX market:

- The current FX spot rate is used as the starting point of the simulation.
- The forward FX curve implied by domestic and foreign interest rate curves determines the expected path of the FX rate.
- The volatility parameter is calibrated using historical time series of FX returns.

This alignment with market observables ensures that simulated FX paths are economically meaningful and consistent with current market conditions.

6.3. Representation of Implied Volatility

Option exposures depend on the implied volatility surface of the underlying FX rate. In principle, each point of the volatility surface could evolve independently as a separate stochastic risk factor. However, modelling the full dynamics of a volatility surface introduces substantial complexity and requires strong assumptions to enforce arbitrage-free relationships between different points on the surface.

To balance model realism and tractability, the model represents the evolution of the volatility surface using a single volatility shift factor. Under this assumption, all points of the volatility surface move proportionally:

$$S_k(\Psi_i, t) = Y_k^{IV}(t) S_k(\Psi_i, 0) \quad (25)$$

This modelling choice is supported by empirical observations that implied volatilities across maturities and deltas tend to move in the same direction during market shocks.

The volatility shift factor is modelled using a mean-reverting stochastic process:

$$dX_k^{IV}(t) = -\lambda_k X_k^{IV}(t)dt + \sigma_k dW_k(t) \quad (26)$$

The use of mean reversion reflects the stylized behaviour of volatility observed in financial markets, where periods of elevated volatility tend to revert toward long-run average levels.

6.4. Calibration Methodology

The model parameters are calibrated using a combination of market data and historical observations.

The deterministic component of the FX process is derived directly from observable interest rate curves. Volatility parameters are estimated using historical FX time series, typically over a multi-year window. This approach provides a statistically robust estimate of FX return variability.

For the volatility shift factor governing implied volatility dynamics, parameters are estimated using historical time series of volatility surface movements. However, due to instability observed in the optimisation procedure, conservative parameter values are adopted for the volatility and mean reversion speed. This choice prioritizes model stability and robustness over exact calibration to historical data.

6.5. Correlation Structure

The model incorporates correlations between risk factors through the correlation matrix of the stochastic drivers. These correlations are estimated from historical residuals of the model processes.

The use of residual correlations ensures that the dependency structure between risk factors reflects the co-movements observed in historical market data. Certain simplifying assumptions are adopted, including the assumption that volatility shift factors are uncorrelated with other risk factors. This assumption reduces model complexity and reflects the limited empirical evidence of stable correlations between volatility surface movements and other market factors.

6.6. Strengths of the Modelling Framework

The FX simulation model presents several conceptual strengths:

- The model is grounded in well-established financial theory and ensures consistency with interest rate parity.
- The geometric Brownian motion specification ensures that FX rates remain strictly positive.
- The modelling approach is computationally efficient and suitable for large-scale Monte Carlo simulation frameworks.
- The simplified representation of the volatility surface reduces dimensionality while preserving the main dynamics observed in implied volatility movements.

These characteristics make the model appropriate for exposure simulation and counterparty credit risk applications.

6.7. Model Limitations

Despite its strengths, the modelling framework involves several simplifying assumptions.

First, the geometric Brownian motion assumption implies constant FX volatility and does not capture potential features such as volatility clustering or regime changes observed in FX markets.

Second, the simplified volatility surface model assumes that all points on the surface move proportionally. This assumption does not capture more complex surface deformations such as twists, skews, or localised volatility shocks.

Third, correlations between risk factors are estimated using historical data and assumed to remain stable over time. In stressed market environments, correlations may deviate significantly from historical averages.

Finally, the use of global constant parameters for volatility shift factors may limit the model's ability to fully reproduce the dynamics of specific volatility surfaces.

6.8. Overall Assessment

Overall, the FX simulation model is conceptually sound for its intended application within the exposure simulation framework. The modelling choices strike a balance between theoretical consistency, empirical realism, and computational tractability.

While the model relies on simplifying assumptions, these assumptions are reasonable given the primary objective of the framework, which is to generate realistic risk factor scenarios for portfolio exposure calculations. The model structure is consistent with standard practices in quantitative risk modelling and is appropriate for large-scale Monte Carlo simulation environments.

7. Implementation

This section provides an overview of the numerical implementation of the FX simulation model within the exposure simulation framework.

7.1. Simulation Framework

The FX model is implemented within a Monte Carlo simulation engine designed to generate future scenarios of market risk factors. The simulation framework jointly evolves multiple risk factors, including interest rates, FX rates, credit spreads, inflation indices and volatility factors.

Each risk factor is represented by a stochastic process driven by correlated Brownian motions. The joint evolution of the processes is obtained through the application of a Cholesky decomposition of the risk factor correlation matrix, ensuring that simulated shocks reproduce the dependency structure observed in historical data.

7.2. Integration with the Risk Engine

The FX model is integrated into the broader risk factor simulation engine used for exposure calculations. At each simulation time step, the model generates FX rate scenarios which are subsequently used in the revaluation of FX-dependent instruments within the portfolio.

The implementation is designed to be computationally efficient and scalable, allowing the simulation of a large number of market scenarios required for counterparty credit risk metrics such as Potential Future Exposure (PFE), Expected Exposure (EE), and Expected Positive Exposure (EPE).

7.3. Reproducibility and Controls

The implementation includes standard controls to ensure reproducibility and numerical stability. These controls include fixed random number seeds for testing environments, monitoring of simulation outputs, and consistency checks between simulated market variables and their deterministic components.

The numerical implementation has been designed to ensure stability, efficiency, and consistency with the theoretical model specification described in the previous sections.

8. Performance

This section evaluates the empirical adequacy of the FX simulation model by comparing its underlying assumptions with the statistical properties of historical FX returns. A validation prototype was implemented in Python to test each major modelling assumption through a dedicated diagnostic script.

The objective of this analysis is not to validate pricing accuracy but to determine whether the simplified modelling assumptions remain acceptable for the intended use of the model, namely the generation of FX scenarios for Potential Future Exposure (PFE) calculations.

8.1. Performance Testing Framework

The validation prototype consists of a suite of scripts designed to assess the key assumptions of the model:

- descriptive statistics of FX log-returns;
- rolling volatility diagnostics to evaluate the constant volatility assumption;
- tail diagnostics including skewness, excess kurtosis, and extreme value indicators;
- statistical tests of the Gaussian innovation assumption;
- exception-rate backtesting of model-implied tail quantiles;
- regime detection using unsupervised clustering methods.

Each script corresponds to a specific modelling assumption and produces reproducible diagnostics for empirical assessment.

8.2. Return Distribution Diagnostics

The analysis of historical FX log-returns indicates that the return distribution deviates from the normal distribution assumed in the simulation model. In particular, empirical return distributions typically exhibit:

- excess kurtosis relative to the Gaussian benchmark;
- occasional large return realizations significantly exceeding the range predicted by a normal distribution;
- moderate skewness depending on the currency pair and sample period.

These characteristics suggest that extreme FX movements occur more frequently than predicted by the lognormal diffusion assumption. As a result, the model may underestimate tail risks when used to generate exposure distributions.

8.3. Volatility Stability

Rolling volatility diagnostics reveal significant time variation in realized FX volatility. Periods of relatively stable volatility are frequently interrupted by episodes of heightened market stress during which realized volatility increases materially.

This behaviour is consistent with the well-documented phenomenon of volatility clustering in financial time series. Since the simulation model assumes constant volatility calibrated over a fixed historical window, it cannot fully capture such regime-dependent volatility dynamics.

Consequently, simulated FX paths may understate the persistence of volatility shocks observed in historical data.

8.4. Tail Behaviour

Tail diagnostics based on QQ-plot analysis and extreme value indicators provide further evidence of heavy-tailed behaviour in FX returns. In particular:

- the empirical distribution exhibits heavier tails than the normal distribution;
- extreme return realizations occur more frequently than implied by the Gaussian assumption;
- tail behaviour may vary significantly across market regimes.

These findings indicate that the Gaussian innovation assumption used in the model may lead to an underestimation of extreme FX movements.

8.5. Distributional Tests

Formal statistical tests applied to standardized returns typically reject the hypothesis that the residual distribution is strictly Gaussian. Even after adjusting returns for time-varying volatility using rolling estimates, deviations from normality remain observable.

This suggests that departures from the Gaussian assumption cannot be explained solely by volatility clustering and may reflect structural properties of FX returns, such as fat tails or asymmetric shock distributions.

8.6. Quantile Backtesting

Quantile backtesting performed using historical returns and model-implied Gaussian thresholds indicates that tail exceedances occur slightly more frequently than predicted by the theoretical distribution.

Although such exceedances remain relatively infrequent, their presence suggests that the Gaussian diffusion framework may slightly underestimate the frequency of extreme negative returns.

From a PFE perspective, this may translate into a modest underestimation of extreme exposure outcomes during stressed market conditions.

8.7. Regime Behaviour

Clustering analysis based on rolling volatility and drawdown indicators reveals the presence of distinct volatility regimes in historical FX data. In particular, the data typically exhibits:

- low-volatility periods corresponding to stable market environments;
- intermediate regimes characterized by moderate fluctuations;
- stressed regimes associated with elevated volatility and large return realizations.

The simulation model does not explicitly incorporate regime-switching dynamics and therefore cannot reproduce this behaviour.

8.8. Implications for Model Adequacy

Taken together, the empirical diagnostics highlight several limitations of the FX simulation model:

- the assumption of constant volatility does not fully reflect the time-varying volatility observed in FX markets;
- the Gaussian innovation assumption understates the probability of extreme return realizations;
- the model does not capture volatility regimes or persistent stress periods;
- tail risks may therefore be moderately underestimated in simulated exposure distributions.

However, these limitations must be interpreted in the context of the intended use of the model. The model is designed to provide a simplified representation of FX dynamics for exposure simulation rather than a full structural representation of FX market behaviour.

Provided that these limitations are acknowledged and appropriately monitored, the model may remain acceptable for PFE applications.

8.9. Overall Assessment

The empirical validation prototype confirms that while the FX simulation model captures the general magnitude of FX return variability, it simplifies several important stylized facts of FX markets, including volatility clustering, heavy tails, and regime-dependent behaviour.

These limitations should therefore be clearly documented and considered when interpreting model outputs, particularly in the context of extreme exposure scenarios.

8.10. Summary of Model Limitations Identified by Performance Tests

The empirical diagnostics described above highlight several limitations of the FX simulation model relative to observed FX market behaviour. Table 8 summarises the main modelling assumptions, the evidence obtained from the validation tests, and the implications for exposure simulation.

Model Assumption	Validation Evidence	Identified Limitation	Implication for PFE Simulation
Lognormal FX dynamics	Historical return diagnostics show occasional large return realizations and excess kurtosis relative to the Gaussian benchmark.	The lognormal diffusion assumption does not fully capture the empirical distribution of FX returns, particularly during stressed market conditions.	Extreme FX moves may occur more frequently than implied by the model, potentially leading to an underestimation of tail exposure scenarios.
Constant volatility	Rolling volatility analysis shows significant time variation in realized volatility across market periods.	The model assumes constant volatility calibrated over a fixed historical window and therefore cannot reproduce volatility clustering.	Simulated FX paths may underestimate the persistence of volatility shocks and the magnitude of exposures during periods of elevated market stress.
Gaussian innovations	Statistical tests on standardized residuals reject the hypothesis of strict normality.	The Gaussian innovation assumption does not fully capture the heavy-tailed nature of FX return distributions.	The probability of extreme FX shocks may be underestimated in the simulation framework.
Thin tails (normal distribution)	Tail diagnostics including QQ-plots and EVT indicators reveal heavier tails than predicted by the normal distribution.	The model does not explicitly incorporate heavy-tail dynamics.	Exposure distributions generated by the model may slightly understate extreme PFE outcomes.
Stationary dynamics	Clustering analysis reveals distinct volatility regimes corresponding to calm and stressed market environments.	The model does not incorporate regime-switching dynamics or volatility state dependence.	The simulation framework may not fully reproduce prolonged periods of elevated volatility observed historically.
Tail quantile accuracy	Exception-rate backtesting shows occasional exceedances of model-implied quantile thresholds.	The model may underestimate the frequency of extreme return realizations relative to Gaussian assumptions.	PFE estimates during stressed market conditions may be modestly optimistic.

Summary of FX simulation model limitations identified through empirical performance testing

Overall, the empirical diagnostics confirm that the FX simulation model provides a simplified representation of FX dynamics. While the model captures the general magnitude of FX return variability, it does not fully reproduce several well-documented stylized facts of FX markets, including volatility clustering, heavy tails, and regime-dependent behaviour.

These limitations are inherent to the simplified lognormal diffusion framework adopted for exposure simulation. However, given the intended use of the model for PFE generation rather than front-office pricing, the model may remain acceptable provided that these limitations are appropriately documented, monitored, and considered when interpreting exposure results.

9. Governance

Governance ensures that the model remains fit for purpose for PFE exposure simulation, that calibration inputs and outputs are controlled and reproducible, and that model changes are subject to formal review.

9.1. Model Ownership

The model is managed under the standard three-lines-of-defense framework.

- **Model Owner (First Line of Defense)** is responsible for model design, implementation, documentation, calibration, and ongoing maintenance.
- **Market Data Team** is responsible for sourcing and validating FX spot histories and interest rate curves used in calibration.
- **EMRM Validation (Second Line of Defense)** is responsible for independent validation, challenge, and periodic review of the model.

9.2. Approved Model Scope

The model is approved for the simulation of FX spot rates versus USD within the Monte Carlo PFE engine. Non-USD cross rates are not modelled as independent stochastic risk factors; where required, they are derived deterministically from the simulated USD currency legs.

The model is not approved for front-office pricing, for products requiring stochastic volatility or smile dynamics, or for applications requiring explicit modelling of cross-currency basis or independently simulated non-USD crosses.

9.3. Calibration Governance

Model parameters are calibrated using validated market data and controlled historical time series. Calibration procedures are documented, reproducible, and subject to input, output, and configuration controls. Any manual override or methodological change must follow formal governance and approval procedures.

9.4. Monitoring and Change Management

The model is subject to periodic monitoring of calibration stability, data quality, and performance for its intended PFE use. Material changes to methodology, calibration settings, or implementation must follow formal change control and, where required, independent re-validation.

10. Conclusion

Risk Dimension	Rating	Reasoning
Conceptual Soundness	Amber	The FX simulation model is based on a geometric Brownian motion framework with deterministic drift derived from interest rate curves and volatility calibrated from historical FX returns. The specification is simple, transparent, and consistent with standard exposure simulation practice. However, the conceptual review identified material modelling limitations: the assumption of constant volatility does not capture volatility clustering or regime shifts; Gaussian innovations do not reproduce fat tails and excess kurtosis observed in FX returns; jumps and discontinuous FX moves are excluded; and the interaction between FX and interest rates is simplified through deterministic drift inputs. The implied volatility framework relies on a single-factor proportional shift that cannot reproduce smile, skew, or surface twist dynamics. These limitations are important but acceptable for PFE scenario generation, which supports an Amber assessment.
Implementation	Green	The implementation of the model is considered acceptable. The simulation framework is straightforward, and aligned with the theoretical specification. The numerical approach is standard for Monte Carlo exposure simulation, and no material implementation weaknesses were identified that would prevent its use for the intended purpose. The implementation remains subject to the limitations of the chosen modelling framework, but these are model design limitations rather than coding or numerical defects.
Data Integrity	Green	The model relies on historical FX spot rates, current FX spot levels, and domestic and foreign discount curves. Based on the review performed, input data are of adequate quality and appropriate data checks are in place. Data preprocessing, completeness checks, and controls on curve and spot inputs are sufficient for the intended use of the model. No material data integrity issue was identified.
Performance	Green	The performance assessment indicates that the model is acceptable for exposure simulation, but with recognised limitations. Empirical diagnostics suggest that historical FX returns exhibit volatility clustering, fat tails, and regime-dependent behaviour, which are not fully captured by the constant-volatility Gaussian GBM framework. As a result, the model may underestimate extreme FX moves and high exposure quantiles in stressed conditions. Nevertheless, the model remains adequate for its intended PFE use provided that these limitations are clearly documented and supported by sensitivity testing, ongoing monitoring, and appropriate interpretation of tail exposure measures.
Governance	Green	Governance is considered acceptable. The model scope, ownership, calibration process, and change control framework are sufficiently defined for its intended use in PFE. The documentation now covers assumptions, limitations, data, implementation, performance testing, and monitoring expectations. While the model remains subject to simplifications that require explicit limitation statements and ongoing monitoring, the governance framework is adequate to support controlled use of the model within the institution's model risk management framework.
Overall	Green	Overall, the model is considered acceptable for its intended use in PFE exposure simulation. Its strengths lie in its transparency, simplicity. Its weaknesses are mainly structural and relate to the inability of the model to capture stochastic volatility, jumps, fat tails, regime changes, dynamic correlation effects, and rich implied volatility surface behaviour. These limitations may affect tail exposure accuracy, particularly for portfolios with optionality or during stressed market conditions, but they are understood. They should be however fully and further documented.

Conclusion summary by validation dimension

The model is therefore considered suitable for use in the PFE simulation framework subject to the limitations described in this report.

A. Empirical Validation Prototype for the FX Simulation Model

This appendix documents the empirical validation prototype developed to assess the statistical properties of FX returns relative to the assumptions embedded in the FX simulation model. The prototype consists of a collection of Python scripts designed to evaluate specific modelling assumptions through reproducible diagnostics.

The objective of the prototype is to provide an empirical basis for the performance assessment discussed in the main body of the report.

A.1. Overview of the Validation Framework

The validation prototype is structured as a sequence of diagnostic tests, each corresponding to a key modelling assumption of the FX simulation model. The framework includes the following components:

- computation of FX log-return statistics;
- analysis of rolling realized volatility;
- tail diagnostics including QQ-plot analysis and extreme value indicators;
- statistical tests of the Gaussian innovation assumption;
- quantile exception backtesting;
- clustering-based regime detection.

Each component is implemented in a dedicated Python script to ensure modularity and reproducibility of the validation analysis.

A.2. Data Preparation and Return Construction

Historical FX spot rates are loaded from the market data repository and converted into log-returns according to

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right).$$

Log-returns are used rather than simple returns because the FX simulation model assumes lognormal dynamics. The return series therefore provides the appropriate empirical benchmark against which the model assumptions are evaluated.

A.3. Descriptive Return Diagnostics

The first validation step consists of computing descriptive statistics of the FX log-return series. The statistics include:

- sample mean;
- sample standard deviation;
- minimum and maximum return realizations.

These diagnostics provide a preliminary characterization of the return distribution and highlight the presence of large return realizations that may challenge the Gaussian diffusion assumption.

This analysis is implemented in the script:

01_returns_and_basic_stats.py.

A.4. Rolling Volatility Stability

The FX simulation model assumes constant volatility estimated from historical data over a fixed calibration window. To assess this assumption, realized volatility is computed using rolling windows of different lengths.

Specifically, rolling standard deviations are computed over the following horizons:

- 20 trading days;
- 60 trading days;
- 120 trading days;
- 252 trading days.

The resulting volatility series allow visual inspection of volatility clustering and structural shifts in market conditions.

This analysis is implemented in the script:

02_rolling_vol_stability.py.

A.5. Tail Diagnostics

To assess the adequacy of the Gaussian innovation assumption, the validation framework includes several diagnostics focusing on the tails of the return distribution.

The following measures are computed:

- sample skewness;
- excess kurtosis;
- quantile–quantile plots versus the normal distribution;
- extreme-value threshold diagnostics based on the mean-excess function.

These diagnostics help identify heavy-tailed behaviour or asymmetries in the return distribution that are inconsistent with the normal distribution assumed in the model.

This analysis is implemented in the script:

03_tail_diagnostics_qq_evt.py.

A.6. Distributional Tests of Standardized Residuals

Under the model assumptions, returns standardized by volatility should follow an approximately Gaussian distribution.

To test this implication, returns are standardized using a rolling volatility estimate, and the resulting residuals are tested against the standard normal distribution using several goodness-of-fit tests:

- Kolmogorov–Smirnov test;
- Anderson–Darling test;
- Cramér–von Mises test.

These tests provide formal statistical evidence on whether the standardized residuals are consistent with the Gaussian assumption.

This analysis is implemented in the script:

`04_distribution_tests_standardized_residuals.py`.

A.7. Quantile Backtesting

Since the model is used to generate exposure scenarios, the accuracy of tail quantiles is particularly relevant.

The validation framework therefore includes a rolling quantile backtesting procedure analogous to a Value-at-Risk backtest. Using rolling estimates of the mean and volatility, model-implied quantile thresholds are computed and compared with realized returns.

The frequency of quantile exceedances is then evaluated relative to the theoretical expectation.

This analysis is implemented in the script:

`05_exception_rate_backtest_var_bands.py`.

A.8. Regime Detection

The FX simulation model assumes stationary dynamics with constant volatility. However, financial markets often exhibit regime-dependent behaviour.

To assess the presence of volatility regimes, the prototype includes two clustering approaches:

- K-means clustering applied to rolling volatility and drawdown indicators;
- Gaussian mixture modelling applied to rolling volatility features.

These unsupervised learning methods identify clusters in the historical data corresponding to different market regimes.

The analyses are implemented in the scripts:

- `06_regime_clustering_kmeans.py`;
- `07_regime_clustering_gmm.py`.

A.9. Reproducibility and Execution

All validation diagnostics are implemented as standalone scripts that can be executed independently. Each script produces graphical diagnostics and summary statistics that support the performance assessment described in the main report.

The modular structure of the prototype allows the validation tests to be applied consistently across multiple FX currency pairs and time periods.

Together, the scripts provide a transparent and reproducible empirical framework for assessing the adequacy of the FX simulation model.

B. Documentation of Python Validation Algorithms

This appendix documents the Python prototype developed to assess the empirical adequacy of the FX simulation model. The prototype is structured as a sequence of standalone validation scripts, each designed to test one or more assumptions of the model.

The tests are aligned with the validation objectives of the report, namely:

- stability of calibrated volatility;
- adequacy of Gaussian standardized residual assumptions;
- conservativeness of tail quantiles;
- identification of potential regime effects not captured by the baseline model.

All scripts use a common preprocessing workflow in which an FX series is loaded and transformed into log-returns. The scripts rely on a shared utility layer for loading data, computing returns, and exporting results.

B.1. Common Data Preparation Layer

All validation scripts begin by loading an FX time series through a common helper function and transforming spot observations into log-returns. The scripts import a common interface containing the data loader, the log-return calculator, the output directory manager, and table-writing utilities. This common interface is used in each of the scripts documented below.

The generic return transformation is of the form

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right),$$

which is consistent with the lognormal diffusion assumption used in the FX simulation model.

B.2. Algorithm 1: Returns and Basic Statistics

B.2.0.1 Script

01_returns_and_basic_stats.py

B.2.0.2 Objective

This script provides a first descriptive summary of the historical FX return series. It is intended to establish the sample period, the scale of daily return volatility, and the presence of extreme return realizations.

B.2.0.3 Algorithm

The script:

1. loads the FX spot time series;
2. computes log-returns;

3. extracts the return vector r ;
4. computes the sample size, start date, end date, sample mean, sample standard deviation, minimum return, and maximum return;
5. writes both the summary statistics and the return series to output files.

The implemented statistics are explicitly:

- $n = \text{len}(r)$,
- sample mean $\bar{r}$,
- sample standard deviation using `ddof=1`,
- sample minimum and maximum.

This is directly visible in the script, which computes `mean_daily`, `std_daily`, `min_daily`, and `max_daily` from the return vector and stores them in a summary table :contentReference[oaicite:2]index=2.

B.2.0.4 Outputs

- `stats.csv`
- `returns_series.csv`

B.2.0.5 Validation use

This algorithm provides the baseline empirical characterization of the return series and helps identify whether the observed distribution already suggests features inconsistent with the Gaussian GBM framework, such as unusually large extreme moves.

B.3. Algorithm 2: Rolling Volatility Stability

B.3.0.1 Script

`02_rolling_vol_stability.py`

B.3.0.2 Objective

This script evaluates the stability of realized FX volatility over time and therefore directly tests the plausibility of the constant-volatility assumption used in the simulation model.

B.3.0.3 Algorithm

The script:

1. loads the FX time series;
2. computes log-returns;
3. computes rolling sample standard deviations of returns over four windows: 20, 60, 120, and 252 observations;

4. stores the resulting rolling volatility series;
5. plots selected rolling volatility curves through time.

The script explicitly computes

$$\hat{\sigma}_{t,w} = \text{Std}(r_{t-w+1}, \dots, r_t)$$

for $w \in \{20, 60, 120, 252\}$ using rolling sample standard deviations :contentReference[oaicite:3]index=3.

B.3.0.4 Outputs

- rolling_vol.csv
- rolling_vol.png

B.3.0.5 Validation use

If realized volatility is materially time-varying, clustered, or exhibits structural breaks, the constant-volatility assumption becomes only a coarse approximation. This algorithm therefore provides direct evidence relevant to calibration stability and break detection, both of which are explicitly required by the report's performance and governance framework :contentReference[oaicite:4]index=4.

B.4. Algorithm 3: Tail Diagnostics, QQ Analysis, and EVT Threshold Scan

B.4.0.1 Script

03_tail_diagnostics_qq_evt.py

B.4.0.2 Objective

This script tests whether the empirical FX return distribution exhibits tail behaviour inconsistent with the Gaussian innovation assumption.

B.4.0.3 Algorithm

The script:

1. loads the FX time series;
2. computes log-returns;
3. calculates excess kurtosis and skewness;
4. generates a QQ plot of returns against the normal distribution;
5. computes an EVT-style threshold scan using the mean-excess function on absolute returns.

More precisely:

- excess kurtosis is computed using `scipy.stats.kurtosis` with Fisher's definition;
- skewness is computed using `scipy.stats.skew`;

- the QQ plot is produced using `stats.probplot`;
- the tail scan uses thresholds defined by quantiles from 0.90 to 0.995 of $|r|$, and for each threshold u computes

$$e(u) = \mathbb{E}[|r| - u \mid |r| > u].$$

These calculations are explicitly implemented in the script, including the mean-excess construction based on exceedances above quantile thresholds of absolute returns :contentReference[oaicite:5]index=5.

B.4.0.4 Outputs

- `tail_stats.csv`
- `qqplot_normal.png`
- `evt_threshold_scan.csv`
- `mean_excess.png`

B.4.0.5 Validation use

This algorithm is designed to corroborate whether FX returns exhibit excess kurtosis, heavy tails, or non-normal quantile structure. Such evidence directly supports limitations already identified in the report concerning Gaussian innovations and the potential underestimation of tail exposure.

B.5. Algorithm 4: Distribution Tests on Standardized Residuals

B.5.0.1 Script

`04_distribution_tests_standardized_residuals.py`

B.5.0.2 Objective

This script tests whether returns standardized by rolling volatility are approximately Gaussian, as implied by the model once volatility scaling is taken into account.

B.5.0.3 Algorithm

The script:

1. loads the FX time series;
2. computes log-returns;
3. estimates rolling volatility using a 60-observation window;
4. standardizes returns according to

$$z_t = \frac{r_t}{\hat{\sigma}_t};$$

5. applies three distributional tests to the standardized residuals:

- Kolmogorov–Smirnov test versus $N(0, 1)$,
- Anderson–Darling test for normality,
- Cramér–von Mises test versus the standard normal CDF.

The use of a 60-day rolling standard deviation and the three formal tests is explicitly coded in the script, which outputs the KS statistic and p -value, the CvM statistic and p -value, and the AD statistic with its critical values :contentReference[oaicite:6]index=6.

B.5.0.4 Outputs

- `tests_summary.csv`
- `standardized_residuals.csv`

B.5.0.5 Validation use

This algorithm directly implements the type of standardized residual testing recommended in the report's performance section, which specifically refers to KS, AD, and CvM testing on standardized residuals / PIT-style diagnostics :contentReference[oaicite:7]index=7.

B.6. Algorithm 5: Exception-Rate Backtest of Gaussian VaR Bands

B.6.0.1 Script

`05_exception_rate_backtest_var_bands.py`

B.6.0.2 Objective

This script evaluates whether a Gaussian volatility-based tail threshold produces an empirically reasonable frequency of exceptions. It serves as a proxy conservativeness test for tail modelling.

B.6.0.3 Algorithm

The script:

1. loads the FX time series;
2. computes log-returns;
3. estimates a rolling mean and rolling standard deviation using a 60-observation window;
4. computes a one-sided 99% Gaussian return threshold

$$\text{VaR}_{0.99,t} = \hat{\mu}_t + z_{0.01} \hat{\sigma}_t,$$

where $z_{0.01}$ is the 1% quantile of the standard normal distribution;

5. flags a breach whenever

$$r_t < \text{VaR}_{0.99,t};$$

6. counts the number of breaches and compares it with the expected number $0.01 \times n$;
7. applies a two-sided binomial test to assess whether the observed breach count is statistically consistent with the expected frequency.

All of these steps are explicitly visible in the script, including the rolling mean, rolling volatility, breach indicator, breach count, and binomial test `stats.binomtest :contentReference[oaicite:8]index=8`.

B.6.0.4 Outputs

- `summary.csv`
- `series.csv`

B.6.0.5 Validation use

This algorithm implements the exception-rate logic requested in the report for selected tail quantiles and therefore supports the assessment of whether the model may underestimate tail frequency or produce optimistic tail thresholds.

B.7. Algorithm 6: Regime Clustering by K-Means

B.7.0.1 Script

`06_regime_clustering_kmeans.py`

B.7.0.2 Objective

This script identifies empirical volatility regimes in the FX series using unsupervised clustering. It is intended to detect whether the return process exhibits persistent states inconsistent with the stationary constant-volatility assumption.

B.7.0.3 Algorithm

The script:

1. loads the FX series and computes log-returns;
2. constructs short- and medium-horizon rolling volatility features;
3. constructs an additional stress indicator based on rolling cumulative returns / drawdown-style behaviour;
4. standardizes the feature space;
5. applies K-means clustering with a fixed number of clusters;
6. maps each observation to a cluster interpreted as a volatility regime.

B.7.0.4 Outputs

Typical outputs include a regime-labelled time series and summary tables by cluster.

B.7.0.5 Validation use

The presence of distinct clusters corresponding to calm, transition, and stressed periods would provide evidence that the constant-volatility GBM specification suppresses relevant regime effects in the historical FX process.

B.8. Algorithm 7: Regime Clustering by Gaussian Mixture Model

B.8.0.1 Script

07_regime_clustering_gmm.py

B.8.0.2 Objective

This script provides an alternative regime-detection framework using a Gaussian mixture model. Unlike K-means, it allows soft probabilistic assignment of observations to latent regimes.

B.8.0.3 Algorithm

The script:

1. loads the FX series and computes log-returns;
2. constructs rolling volatility features;
3. standardizes the features;
4. fits a Gaussian mixture model with a fixed number of latent components;
5. computes posterior regime probabilities and assigns the most likely regime to each observation.

B.8.0.4 Outputs

Typical outputs include posterior cluster probabilities and regime assignments through time.

B.8.0.5 Validation use

This algorithm is useful for identifying whether the FX process exhibits latent heterogeneity and whether stressed states are sufficiently persistent to challenge the stationary model assumption.

B.9. Interpretation within the Validation Framework

Taken together, the Python algorithms form a coherent empirical validation suite. Their roles can be summarized as follows:

- Algorithms 1 and 3 assess whether unconditional returns exhibit stylized facts inconsistent with Gaussian GBM assumptions;
- Algorithm 2 evaluates the empirical plausibility of constant volatility;

- Algorithm 4 tests the standardized innovation assumption more directly;
- Algorithm 5 evaluates the conservativeness of Gaussian tail thresholds;
- Algorithms 6 and 7 test for regime effects omitted by the baseline model.

This structure is fully consistent with the report's stated validation priorities of calibration stability, distributional testing, quantile exceedances, and regime-shift detection :contentReference[oaicite:9]index=9.

B.10. Limitations of the Prototype

The current scripts rely on a shared utility module for loading the FX series and exporting outputs. Accordingly, reproducibility depends on access to the same common helper layer and input dataset.

In addition, the clustering algorithms are exploratory tools rather than formal statistical hypothesis tests. Their outputs should therefore be interpreted as supporting diagnostics rather than stand-alone validation decisions.

Finally, the prototype appears to operate on one FX series at a time. For full validation coverage, the framework should be applied systematically to all material FX pairs used in the production PFE engine.